

Speed and velocity

Answers

Calculating average speed

1. The average speed (v) for a journey is calculated using the formula:

$$v = \frac{d}{t}$$

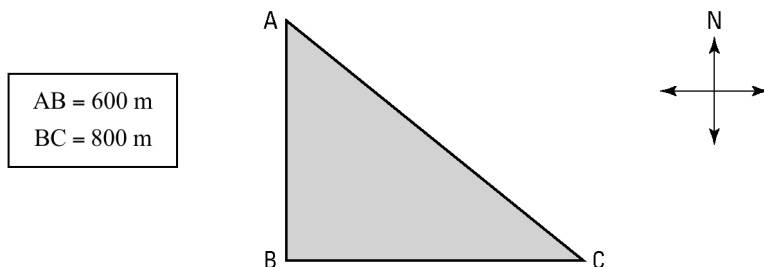
where d = distance travelled and t = time taken.

Complete the following table. Make sure you include the appropriate units for each quantity.

Distance travelled	Time taken	Average speed
100 m	20 s	$\frac{100}{20} = 5 \text{ m/s}$
3 km	$\frac{3}{60} = 0.05 \text{ h}$	60 km/h
$20 \times 15 = 300 \text{ m}$	15 s	20 m/s
2 km	$\frac{2000}{25} = 80 \text{ s}$	25 m/s

Average speed and average velocity

2. Two vehicles (X and Y) travel from A to C along different paths as shown in the diagram (not drawn to scale). X travels along the path ABC and vehicle Y travels along AC. The triangle ABC is a right-angled triangle. A is north of B.



Both vehicles take 70 s for the trip.

Perform the necessary calculations to complete the table.

Average speed of X	Average velocity of X	Average speed of Y	Average velocity of Y
$d = 800 + 600$ $= 1400 \text{ m}$ $t = 70 \text{ s}$ $v = \frac{d}{t} = \frac{1400}{70}$ $= 20 \text{ m/s}$	$v = \frac{\text{displacement}}{\text{time}}$ $= \frac{1000}{70}$ $= 14.3 \text{ m/s}$ Direction = 37° S of E (draw a scale diagram and measure the angle)	$d(\text{AC}) = 1000 \text{ m}$ (this is a 3:4:5 right-angled triangle) $t = 70 \text{ s}$ $v = \frac{d}{t} = \frac{1000}{70}$ $= 14.3 \text{ m/s}$	$v = \frac{\text{displacement}}{\text{time}}$ $= \frac{1000}{70}$ $= 14.3 \text{ m/s}$ Direction = 37° S of E (draw a scale diagram and measure the angle)